Chapter 8 Problem Set

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1. The statement in 8.1 was that $a^{p-1} \equiv 1 \pmod{p}$ and 8.2 was $a^p \equiv a \pmod{p}$. We can deduce the second statement from the first one easily by multiplying both sides of the congruence by a.

We can deduce 8.1 from 8.2 by dividing both sides of the congruence by a. We can do this because $\gcd(a, p) = 1$ since p is prime and $p \not| a$.

2. Base case: We know that $1^p \equiv 1 \pmod{p}$, so it holds true for $a = 1$.

Inductive Step: We can start by assuming that $a^p \equiv (a-1)^p \equiv \cdots \equiv a \pmod{p}$. Our goal is to prove that $(a+1)^p \equiv a+1 \pmod{p}$ Since $(x+1)^p = 1+px+\frac{p(p-1)}{2!}x^2+\cdots+x^p$, we know that $(a+1)^p \equiv 1+pa+\frac{p(p-1)}{2!}+\cdots+pa^{p-1}+x^p \pmod{p}$. We also know that, since p is prime, $p \mid C(p,k)$. That means $(a+1)^p \equiv 1 + a^p \equiv a+1 \pmod{p}$.

3. By Fermat's Little Theorem, $a^{p_i} \equiv a \pmod{p_i}$. Since d is a multiple of all p_i, $a^{d+1} \equiv a \pmod{p_i}$. Since all p_i are relatively prime to each other, by the Chinese Remainder Theorem, $a^{d+1} \equiv a \pmod{p_1 p_2 \dots p_k}$.

4. First of all, $\sum_{i=1}^{p-1} i = \frac{p(p-1)}{2}$. Also, since p is prime, $\frac{p-1}{2}$ is an integer, and $\gcd(p, \frac{p-1}{2}) = 1$. We already know, by Wilson's Theorem, that $(p-1)! \equiv -1 \equiv (p-1) \pmod{p}$. So, by the Chinese Remainder Theorem, if $(p-1)! \equiv (p-1) \pmod{\frac{p-1}{2}}$, then the statement we want to prove holds true.

To prove that, we notice that $2(p-1)! \equiv 0 \equiv 2(p-1) \pmod{p-1}$. And if $ac \equiv bc \pmod{n}$, then $b \equiv c \pmod{\frac{n}{\gcd(a,n)}}$. This means $(p-1)! \equiv (p-1) \pmod{\frac{p-1}{2}}$, as desired.

5. If $(n-1)! \equiv -1 \pmod{n}$, then n is prime.

For the sake of contradiction, let's assume that n is composite. That means it can be written as ab where a and b are positive integers greater than 1 and less than n. By those constraints, a and b must be part of the product $(n-1)!$. However, that means ab divides $(n-1)!$, and thus, $(n-1)! \equiv 0 \pmod{n}$, which is a contradiction.

Now, we need to consider the case where n is a perfect square. Let's say $n = a^2$. We know that $2a \leq n$, and equality only holds when $n = 4$. However, when $n = 4$, $(4-1)! \equiv 2 \not\equiv -1 \pmod 4$. Therefore, we can assume that $2a < n$. That means a and $2a$ are part of the product $(n-1)!$, so $(n-1)!$ is divisible by $2a^2$, and thus, $(n-1)! \equiv 0 \pmod n$, which is a contradiction.

6. By Wilson's Theorem, we know that p divides $(p-1)!+1$. Also, $(p-1)!$ is divisible by every prime smaller than p. That means when we add 1, it is not a multiple of any prime smaller than p. Since p divides $(p-1)!+1$ but no prime smaller does, p is the smallest prime that divides $(p-1)!+1$.

7. Base case ($n = 1$): This means $a^{(p-1)p^0} \equiv a^{p-1} \equiv 1 \pmod{p^1}$ by Fermat's Little Theorem.

Inductive step: Assume that $a^{(p-1)p^{k-1}} \equiv 1 \pmod{p^k}$. We need to prove that $a^{(p-1)p^k} \equiv 1 \pmod{p^{k+1}}$. We can rewrite $a^{(p-1)p^k}$ as $(a^{(p-1)p^{k-1}})^p$. Since $a^{(p-1)p^{k-1}} \equiv 1 \pmod{p^k}$, we can say that $a^{(p-1)p^{k-1}} = 1 + mp^k$. That means $(1+mp^k)^p = 1 + C(p,1)mp^k + C(p,2)m^2p^{2k} + \cdots + (mp^k)^p$. Each term $C(p,i)m^ip^{ik}$ for $i \geq 2$ is divisible by p^{k+1}, so $(1+mp^k)^p \equiv 1 \pmod{p^{k+1}}$, which completes the inductive step.

8. Let $S = \{x_1, x_2, \ldots, x_{\phi(n)}\}$ be the set of positive integers less than n that are relatively prime to n. Also, let T be the set of numbers where each element is the corresponding element in S multiplied by a. Since $\gcd(a,n) = 1$, the elements of T modulo n are just a permutation of those in S modulo n.

Now, let's take the product of all elements in each set: $\prod_{i=1}^{\phi(n)} x_i$ for set S and $\prod_{i=1}^{\phi(n)} ax_i$ for set T. We can rewrite the product for set T as $a^{\phi(n)} \prod_{i=1}^{\phi(n)} x_i$. Since T is a permutation of S modulo n, $a^{\phi(n)} \prod_{i=1}^{\phi(n)} x_i \equiv \prod_{i=1}^{\phi(n)} x_i \pmod n$. We know that $\gcd(\prod_{i=1}^{\phi(n)} x_i, n) = 1$ because every term in the product is relatively prime to n. Therefore, $a^{\phi(n)} \equiv 1 \pmod n$.

9. We are given that $6k+1$, $12k+1$, and $18k+1$ are prime. This means $a^{6k} \equiv 1 \pmod{6k+1}$, $a^{12k} \equiv 1 \pmod{12k+1}$, and $a^{18k} \equiv 1 \pmod{18k+1}$. We can equivalently state that $a^{36k} \equiv 1 \pmod{6k+1}$, $a^{36k} \equiv 1 \pmod{12k+1}$, and $a^{36k} \equiv 1 \pmod{18k+1}$. It follows from the Chinese Remainder Theorem that $a^{36k} \equiv 1 \pmod{(6k+1)(12k+1)(18k+1)}$.

Evaluating $n-1$, we get $(6k+1)(12k+1)(18k+1)-1 = 36k(36k^2+11k+1)$. From earlier, we can say that $a^{36k} \equiv a^{36k(36k^2+11k+1)} \equiv a^{n-1} \equiv 1 \pmod n$. Since n is composite and it satisfies $a^{n-1} \equiv 1 \pmod n$, it is a Carmichael number.

10. We can simplify $C(2p, p)$ as $\frac{(2p)!}{p!p!} = 2 \cdot \frac{(2p-1)(2p-2)\cdots(p+1)}{(p-1)(p-2)\cdots(1)}$. We can rewrite that expression as $2 \cdot \frac{(p+(p-1))(p+(p-2))\cdots(p+1)}{(p-1)\cdots(1)}$. If we take that modulo p^2 and expand it, any term containing p^2 wouldn't contribute to the expression modulo p^2. Keeping that in mind, we can see that $C(2p, p) \equiv 2 \cdot \frac{(p-1)(p-2)\cdots(1)}{(p-1)\cdots(1)} \equiv 2 \pmod{p^2}$.